Self-Supervised Latent Representations for Imbalanced Apical Periodontitis Grading

Project 8 · Computer Vision A.A. 2025–2026

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The dataset

Panoramic radiographs, one bounding box and one PAI grade per lesion · Do, H. V. et al.,

Split	Images	Lesions	PAI 3	PAI 4	PAI 5
Train	2,746	4,719	3,017	1,229	473
Validation	588	1,009	617	268	124
Test	590	1,013	643	258	112

No patient appears in two splits

3,924 distinct patient IDs, recovered from the filename field of the XML annotations. Splitting per lesion instead would put two lesions of the same jaw on both sides of the wall; the split is patient-level, and every run re-checks it.

State of the art

I-JEPA Assran et al., CVPR 2023

predicts representations, not pixels — the architecture the assignment prescribes

Attention-based deep MIL Ilse et al., ICML 2018

a lesion is a bag of tokens; embedding- against instance-level

Bag-size bias the known failure mode of MIL

named in the literature, never measured where cardinality is the label

PR over ROC Saito & Rehmsmeier, 2015

fixes our primary metric: PR-AUC on PAI 5

Experimental setup

Windowing	224 px crop at native resolution → 14 × 14 grid → 196 tokens, fixed by construction
Pre-training	ViT-S/16 · context + EMA target + shallow predictor · lr 3e-5, EMA τ 0.9996 · 300 epochs
Downstream	encoder frozen · attention pooling · blocks 2, 7, 11 concatenated · 5 seeds per cell
Primary metric	PR-AUC on PAI 5

1

Build

We pre-train I-JEPA ourselves.

4,719 lesions, no labels, 289 epochs.
No downloaded weights, no fine-tuning.

→

2

Measure

Freeze it and read it as prescribed.

The trained encoder does not separate itself from one with random weights.

→

3

Grading

The patch is graded

The model decides the gravity of lesion by giving it PAI 3/4/5

Two stages, one frozen encoder

self-supervised, no labels

Context encoder

ViT-S/16
12 blocks, dim 384

21.6 M

Predictor

4 blocks — narrow

0.5 M   2.4 %

Target encoder

ViT-S/16
12 blocks, dim 384
EMA only, no gradient

21.6 M

predicts representations, not pixels

the target encoder is then frozen

downstream, supervised

Token selection

Fixed size

Attention pooling

5.3 M

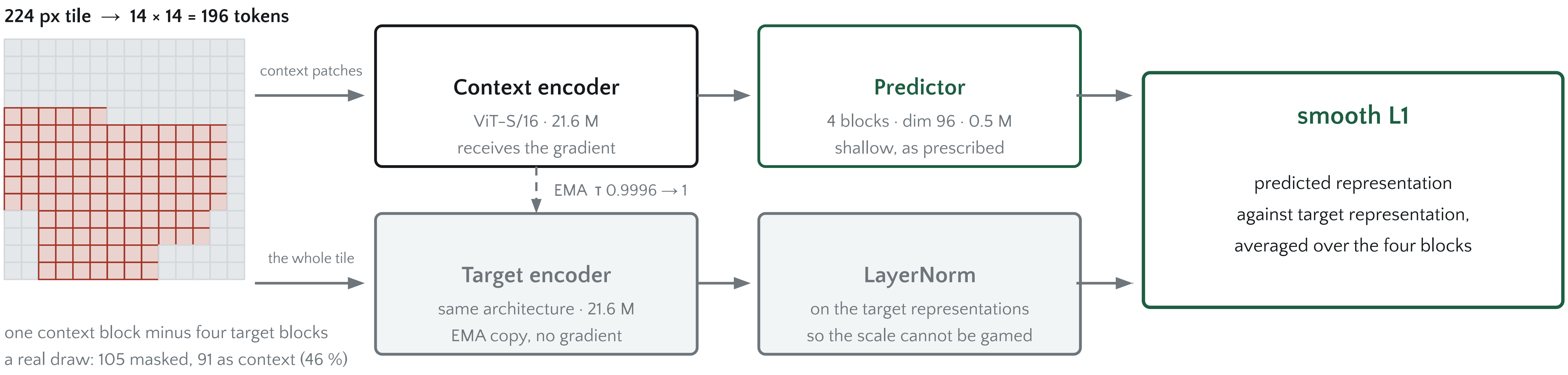
Classifier head

one Linear, 1,152

3,459   0.1 %

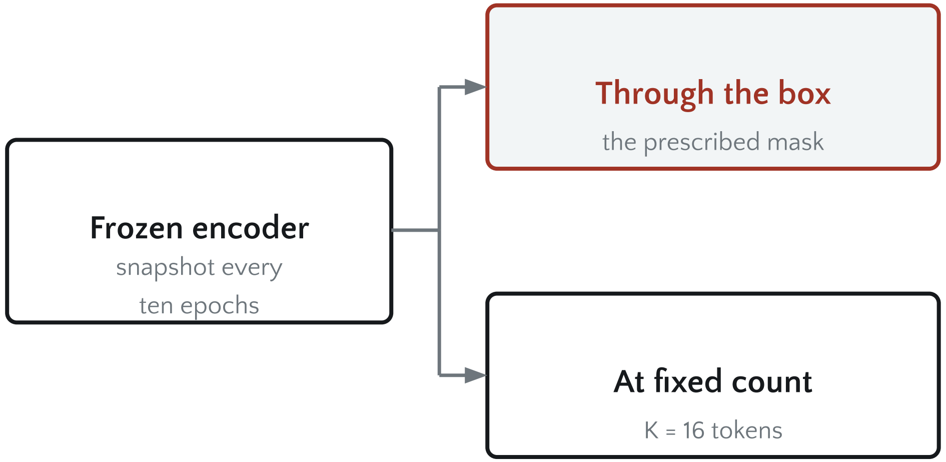
How I-JEPA works

Assran et al., CVPR 2023 · the prediction happens in representation space — no pixel is ever reconstructed

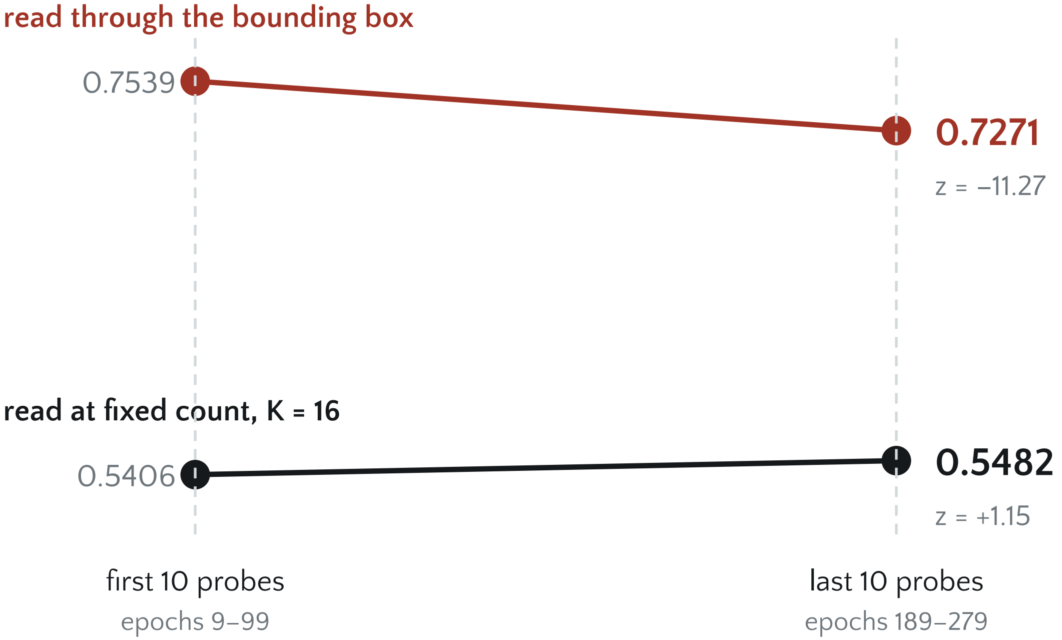


The target is a representation the network produces itself — the loss never asks for texture.

Does the encoder learn?



same head, same seeds, same epochs — only the mask changes

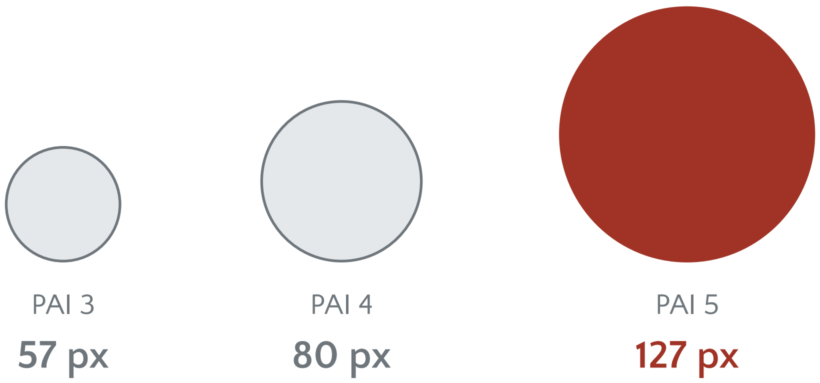


vertical positions to scale, one shared axis

The problem is geometric

Random ViT and I-JEPA, both frozen · as prescribed, bounding-box mask · macro-F1, test, 5 seeds

The PAI grade is, largely, lesion size — and the box hands that size to the classifier.



median bounding-box side, drawn to scale

What the classifier is given	Image	macro-F1
Bounding-box mask only — every pixel zeroed	none	**0.7708**
Random ViT, full latent vector	full	**0.7705**
I-JEPA, full latent vector	full	**0.7663**
Two thresholds on the box side — no network	none	**0.7567**

So the box selects **16 / 36 / 64** tokens for PAI 3 / 4 / 5. The count is nearly the label — and a classifier can read it without looking at a single token.

What the frozen encoder is worth

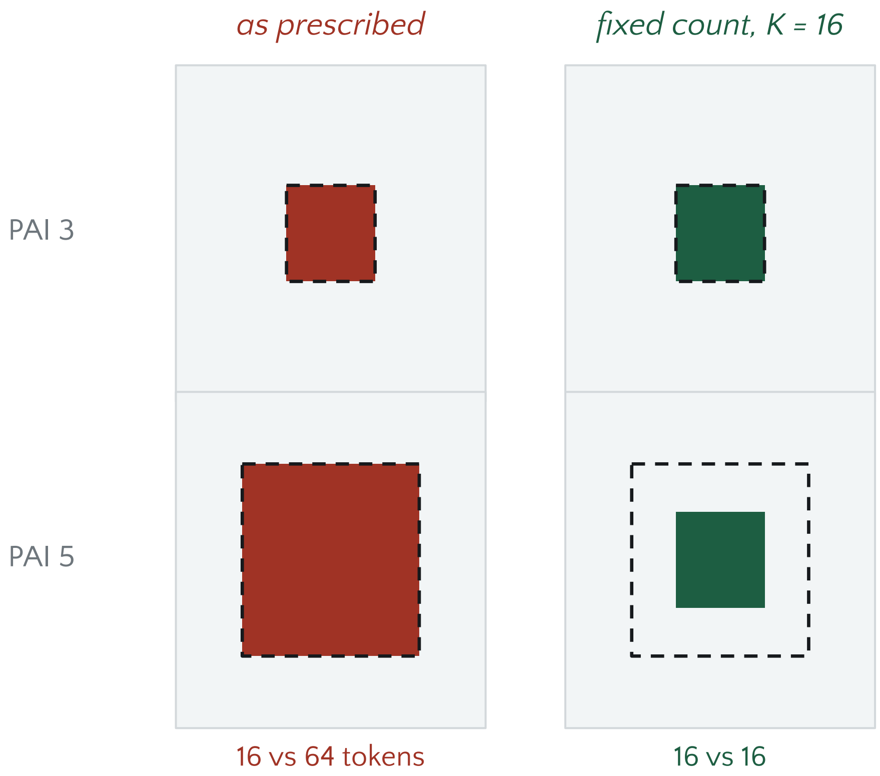
I-JEPA vs random ViT, both frozen · PR-AUC on PAI 5, test, 5 seeds

Protocol	Bag-size cue	Localisation	Random	I-JEPA	Gain over random
P1	present	present	0.8758	0.8785	–
P3 — K = 16	removed	present	0.3861	**0.4713**	**22.1%**
P3 — K = 36	removed	present	0.3823	**0.5031**	**31.6%**
P3 — K = 64	removed	present	0.3740	**0.4553**	**21.7%**
Size-blind crop	removed	present	0.4452	**0.6673**	**49.9%**

Up to +32 % on the rare class by changing only which tokens are aggregated — and +50 % when the crop itself is made blind to size.

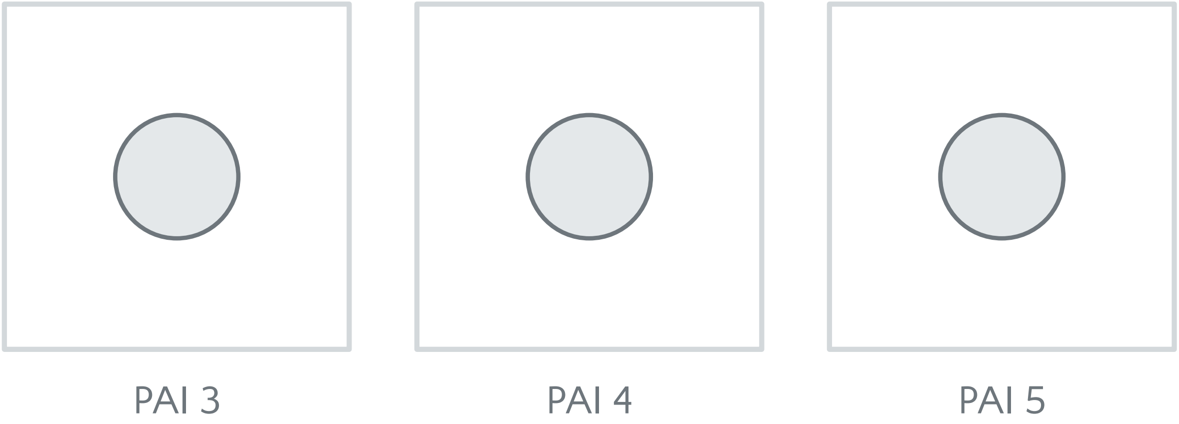
The box becomes a pointer

Keep the box centre. Take the K nearest tokens. Same K for every class.



Size-blind crops: a suitable solution

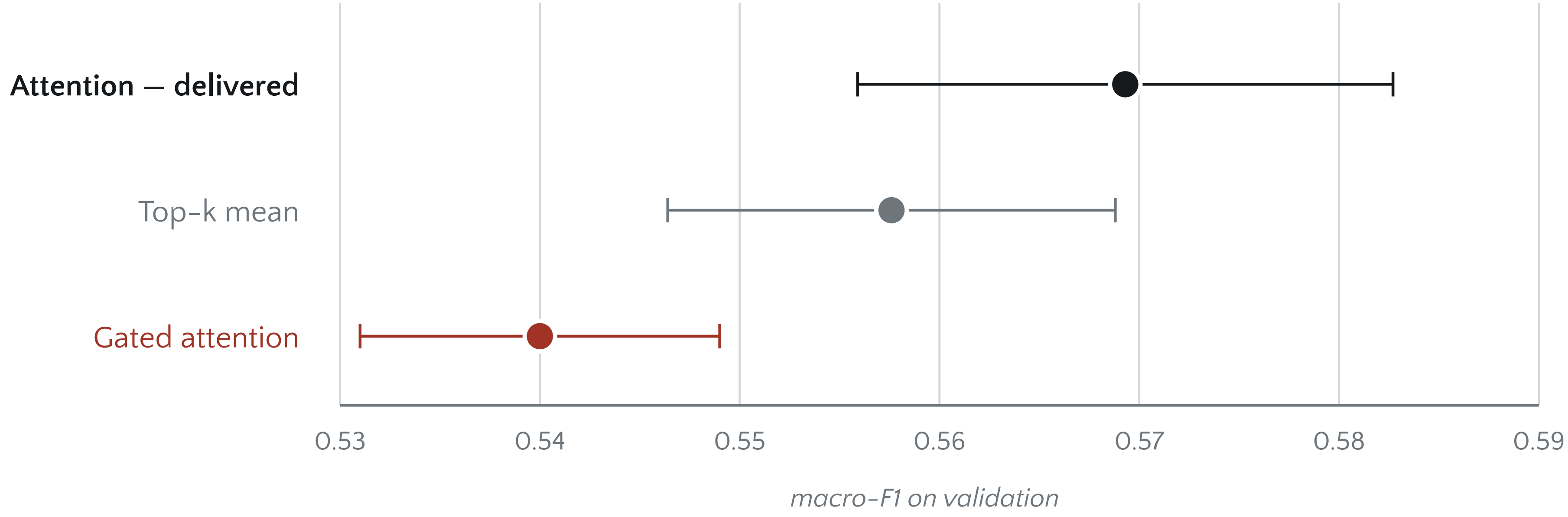
Size-blind cropping proved to be a very promising solution during our initial evaluation . However, stretching these crops to fit a fixed resolution distorts the actual geometry of the lesions. Considering our project timeline, the fact that we are training a small model and the importance of preserving natural anatomical proportions, we decided not to adopt this approach for our main experiments. We still keep it as a control baseline, as it represents one of the most exciting avenues for future deep-dive research.



36 tokens, for every grade

How is pooling performed?

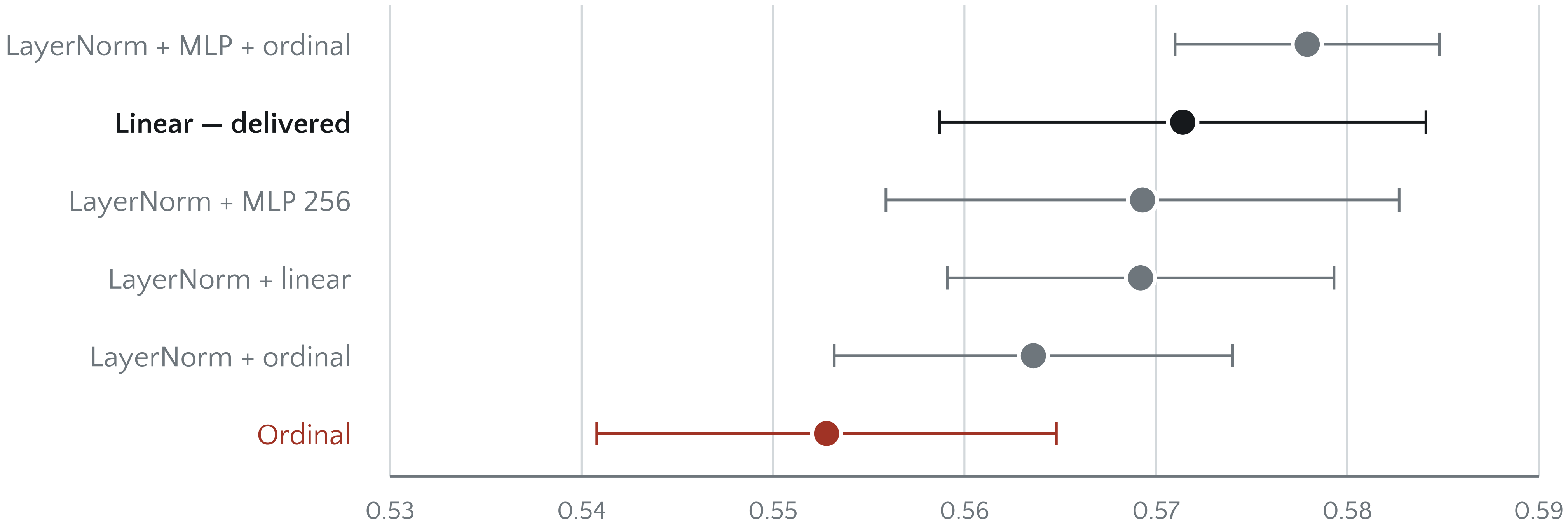
A single learned query scores the K selected tokens. A softmax turns the scores into weights, the tokens are summed with those weights, and one LayerNorm plus one Linear give the three logits. It fits the shape of the problem. The tokens are a bag with no meaningful order, so the aggregator has to be permutation-invariant; and it re-projects the tokens before averaging, so it learns what to read and not only where to look.



Does the head matter?

Six heads on the same frozen I-JEPA encoder · fixed count, K = 16 · macro-F1 on validation, 5 seeds, ± 1 sd between seeds

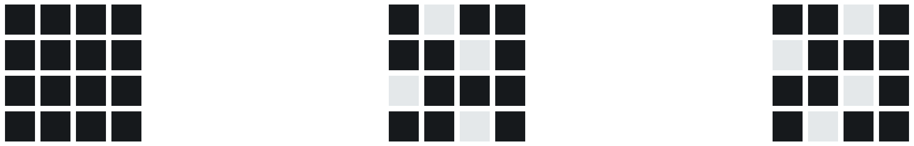
Simplicity Wins: The standard Linear head delivers top-tier performance without adding architectural complexity or extra parameters. The downstream performance is driven primarily by the quality of the latent representations, not head depth.



Balanced token sampling

Any frozen encoder · against focal, class-weighted, oversampling, SMOTE · PR-AUC on PAI 5

For the rare classes, sample different subsets of one lesion's tokens. Each subset is one instance.



view 1 — intact   view 2   view 3

$$n_c = \left(n_{max} / n_c \right)^{\alpha}$$ rounded up

at $\alpha = 0.5$ that is 1 / 2 / 3 views for PAI 3 / 4 / 5

Method	What it multiplies	Invents data
Oversampling	identical copies	no
Focal / class-weighted	nothing — reweights loss	no
SMOTE	synthetic points	yes
Balanced token sampling	genuine views of a lesion	no

Every instance is a genuine view of a real lesion.

Five ways to handle the imbalance

I-JEPA encoder, frozen · fixed count, K = 16 · PR-AUC on PAI 5, test, 5 seeds

Only one of the five separates from doing nothing.

Class-weighted, focal and oversampling all sit inside the noise of plain cross-entropy, against a threshold of $|z| \geq 2.31$. Our method does see 46 % more gradient steps — and that is not the explanation, because at $\alpha = 1.00$ it sees 112 % more and does worse.

Method	PR-AUC PAI 5	vs plain CE	z
Balanced token sampling subsets of one lesion's tokens, each an instance	0.5150 ± 0.0158	**+0.0437**	**+2.93**
Class-weighted cross-entropy the loss weights rare classes more	0.4935 ± 0.0368	+0.0222	+1.05
Focal loss the loss weights easy examples less	0.4751 ± 0.0132	+0.0039	+0.27
Plain cross-entropy the baseline	0.4713 ± 0.0294	—	—
Oversampling the same lesion drawn again, identical every time	0.4375 ± 0.0386	−0.0338	−1.56

What we built, and what it is worth

1 **We pre-trained I-JEPA ourselves, from scratch.**

289 epochs on 4,719 lesions, no downloaded weights and no fine-tuning. The encoder is ours end to end, and it stays frozen everywhere downstream.

2 **We found that the prescribed evaluation could not measure any encoder — and we proved it.**

The bounding-box mask alone, with every pixel zeroed, reaches macro-F1 0.7708. Four different ways of not using the encoder land on the same number.

3 **We corrected it without leaving the brief, and the encoder appeared.**

Keep the box centre, take K tokens: same crop, same tokens, same seeds. +32 % on PAI 5 at $z = 8.72$, the control cell tying as it must — and a second, independent removal agrees at +50 %.

4 **We contributed a method for the imbalance, and it is the only one that separates.**

Balanced token sampling: different subsets of one lesion's tokens, each a training instance. +0.0437 over plain cross-entropy at $z = 2.93$, where focal, class-weighted and oversampling do not clear the threshold.

References

1 Assran, M., Duval, Q., Misra, I. et al. *Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. CVPR 2023.*

2 Ilse, M., Tomczak, J. M., Welling, M. *Attention-based Deep Multiple Instance Learning. ICML 2018.*

3 Do, H. V., Nguyen, T. T. H., Vo, N. T. N. et al. *A Dataset of apical periodontitis lesions in panoramic radiographs. Data in Brief 54:110486, 2024.*

4 Saito, T., Rehmsmeier, M. *The precision-recall plot is more informative than the ROC plot on imbalanced datasets. PLoS ONE 10(3), 2015.*

5 He, K., Gkioxari, G., Dollár, P., Girshick, R. *Mask R-CNN. ICCV 2017.*

Every number in this deck is recomputed from runs/.json by verify_claims.py.*